

Chapter 10

Wettability Techniques to Monitor the Cleanliness of Surfaces

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10.1 BACKGROUND AND INTRODUCTION

In measuring cleanliness one must define what is meant by clean and dirty. Cleanliness is like beauty. Beauty is in the eyes of the beholder, and cleanliness is in the eyes of the user. Therefore, no universal definition exists of “how clean is clean.” Dirty is typically synonymous with contaminated, and a contaminant is “material in the wrong place.” These concepts are a long-studied area of process control, and as such the authors offer this chapter as a revised and expanded treatment of an earlier offering on the topic.¹

In the broad spectrum of contamination control, a major concern is the presence of organic contamination on various inorganic surfaces. This is especially the case in the aerospace and semiconductor industries, where an extremely low level of contamination has a disproportionate impact on yield, reliability, and performance. In order to control surface contamination of materials, a rapid detection method is required that does not adversely affect the surface. Wettability measurements provide a convenient and rapid method for probing the outermost surface of a material. The technique is highly surface specific, generally exceeding the sensitivity of electron spectroscopies and is sensitive to a fraction of a monolayer.

Organic materials commonly found in the laboratory environment are generally hydrophobic in nature (e.g., vacuum-pump oil vapor). This consideration led to the development of the atomizer test where the surface to be tested was sprayed with a fine mist of water. If the deposited drops spread on the surface, it was considered clean. If they remained as droplets, the surface was considered to be contaminated by an organic material.²

For many years the atomizer test was used in the electronics industry to evaluate and monitor various cleaning procedures. It was used primarily in vacuum-tube manufacturing, where there appeared to be no particular problems with this technique, which was really just a means of estimating the wettability of the surface by water. The advent of semiconductor technology, which used very pure silicon as a starting material, changed this point of view. One of the steps in the preparation and cleaning of polished silicon wafers is the treatment with hydrofluoric acid to remove the oxide from the surface. After removal of the oxide on silicon, the resulting silicon hydride surface was hydrophobic, giving a water contact angle of about 50° . The surface could be made hydrophilic by heating it in air or treating it with an aqueous oxidizing solution, known to remove organic contaminants. It was assumed that these oxidizing treatments removed only trace amounts of organic contaminants from the silicon surface. However, the increase in wettability (i.e., decrease in water contact angle) after oxidation appeared larger than would be expected from the removal of small amounts of organic materials. In fact, the heating of the silicon surface in air or its exposure to an oxidizing solution regenerated the hydrophilic silicon oxide on the silicon substrate, by removing the hydrophobic silicon hydride surface.

Surfaces become contaminated at different rates and to different extents. Perfectly clean oxidized metals, ceramics or glass surfaces will be rapidly contaminated on exposure to ambient air, because of their high surface free energies. The primary driving force for surface contamination is a reduction of the surface free energy via adsorption. In fact, it is often observed that the cleaner the part, the quicker it will get dirty.

10.2 FUNDAMENTALS

All surfaces are energetically unfavorable, since they have a positive free energy of formation. A simple explanation may be given by considering the formation of two new surfaces by cleaving a solid. Bonds have to be broken between atoms or molecules on both sides of the cleavage plane, splitting the solid and creating two new surfaces. Breaking the bonds requires work, resulting in a positive contribution to the total free energy of the system. The majority of this added energy generates the surface free energy of the two new surfaces.

The surface free energy may be minimized in several ways. For example, the optimum shape is a sphere having the lowest surface area/volume ratio. This is why liquid droplets in free space are spherical when external forces, such as

gravity, are negligible. The reorganization of molecules or atoms at surfaces can also cause a reduction in the surface free energy.

The adsorption of molecules on high surface energy solids can greatly reduce the surface free energy. The mechanism that generates organic contamination on clean inorganic surfaces is similar to the phenomenon of molecular adsorption. Another example of the spontaneous reduction of the surface free energy is the reduction of the surface tension of water by the adsorption of a surfactant, an abbreviation for “surface active agent.” For liquids, the surface tension is equivalent to the surface free energy. Surfactant molecules are amphiphilic, rendering them surface active at interfaces between water and air, or water and non-polar materials. They are generally poorly soluble in water. Below the critical micellar concentration, the excess concentration of molecules at the water surface, Γ , and the concentration in water, c , are related by the Gibbs adsorption equation, giving the reduction of the surface tension of water, γ_L , as $d\gamma_L = -\Gamma RT \, d \ln c$, where R is the universal gas constant and T is the temperature.

Physical adsorption (physisorption) and chemical adsorption (chemisorption) are the two principal modes of adsorption of molecules on surfaces. The basis of this distinction is the nature of the bonding between the molecules and the surface. In physisorption, the bonding mechanism involves London dispersion forces (non-polar forces) or polar forces. In chemisorption, a chemical bond, involving substantial rearrangement of electron densities in the adsorbate molecules and the substrate molecules, is formed. The nature of this bond may be ionic, covalent, or metallic.

Normal boiling points can be used to illustrate the relative strength of intermolecular attractive forces (Figure 10.1). The London dispersion forces increase as the size of the electron cloud increases, which is correlated with the mass of the molecule as seen in the Group 4 hydrides (CH_4 through

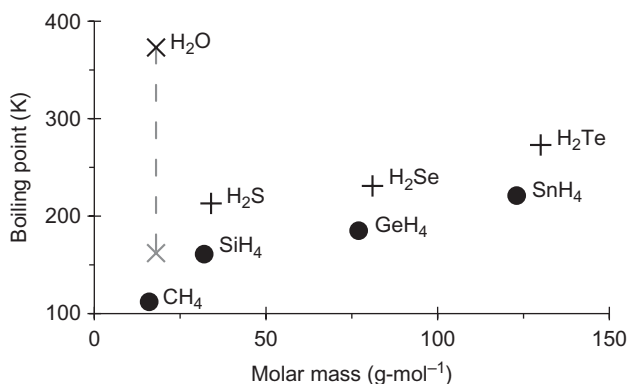


FIGURE 10.1 Boiling points of the Group 4 (●) and Group 6 (+) hydrides show the effects of molecular size (London dispersion forces, CH_4 – SnH_4), polar forces (H_2S – H_2Te), and hydrogen bonding ($\times$, H_2O).



SnH_4 in Figure 10.1). The Group 6 hydrides show an increase in boiling point as polar forces are added. It is clear from this analysis that water does not have a boiling point of 162 K as predicted by polar plus London dispersion forces. Rather, water has the opportunity for H_2O — HOH directional polar bonding. This type of directional bonding is observed when hydrogen is between oxygen, fluorine, and nitrogen atoms, leading to the name hydrogen bonding.

Hydrogen bonding is of particular importance when considering the interactions of water with an oxide-rich surface. This makes water a very sensitive probe of the cleanliness of a metal oxide surface. Any non-hydrogen-bonding substance which blocks the ability of water to form hydrogen bonds with the metal oxide will have a profound impact on the shape of a sessile water drop.

Acid-base interactions are another specific class of intermolecular forces. Lewis acids, which are electron pair acceptors, and Lewis bases, which are electron pair donors, react to form adducts in which a coordinate covalent bond is formed. Acid-base interactions are becoming the subject of growing interest in the field of wetting and adhesion phenomena.^{3–5}

The knowledge of the nature of molecular interactions between contaminants and surfaces is particularly relevant as it can help in the identification of efficient strategies adapted to remove contaminants from a particular surface.

10.3 THEORETICAL AND EXPERIMENTAL INVESTIGATIONS

Whatever the chemical nature of a liquid or a solid, the forces responsible for the cohesion of the material are also responsible for the surface properties of the material. This is specifically the case for surface tension, which results from unbalanced interaction forces at the free surface of a liquid.

A molecule in a liquid is surrounded by other molecules. On average, it is equally attracted in all directions. A molecule on the surface experiences a resultant attraction toward the bulk liquid, due to a greater number of molecules per unit volume in the liquid phase than in the vapor phase. The consequence of this inward pull is that the liquid surface always tends to contract to the smallest possible area.

In order to extend the area of a liquid surface, it is necessary to do work, bringing molecules from the bulk of the liquid to its surface. This work is done against the inward attractive force from the neighboring molecules in the bulk liquid. The work required to increase the area by one unit is called the surface free energy, or surface tension of the liquid. To increase the surface area of the liquid by dA , the work, dW , to be done is

$$dW = \gamma_L dA \quad (10.1)$$

where γ_L is the surface free energy in J/m^2 .

As a result of its tendency to contract, the liquid surface behaves as if it were in a state of tension. The force acting along a length of the liquid surface is proportional to this length, and its value per unit length is called the surface tension

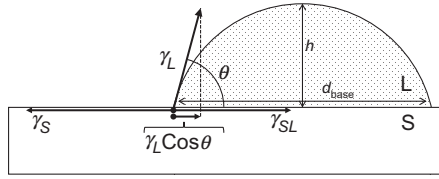


FIGURE 10.2 Profile of a liquid drop on a solid surface.

in N/m. The surface tension of liquids is equal, numerically, to the surface free energy because the conversion of force to work is the same as the conversion of length to area ($\text{N/m} \times \text{m/m} = \text{N} \cdot \text{m/m}^2 = \text{J/m}^2$).

The most widely used quantitative measure of wettability is the contact angle. When a drop of a liquid with a sufficiently small size is placed on a smooth, flat, level, homogeneous solid substrate, the drop takes the shape of a spherical cap (Figure 10.2). The shape of the drop approximates that of a spherical cap when the forces other than the surface tension become negligible. In the case of gravity, the ratio of surface forces to gravitation for a drop is expressed by the capillary length, $\kappa^{-1} = (\gamma_L / \rho g)^{1/2}$, where ρ is the liquid density in g/cm^3 and g is the acceleration due to gravity, 9.81 m/s^2 . A drop whose diameter is substantially smaller than the capillary length will adopt the shape of a spherical cap.

Each solid and liquid combination gives rise to a specific degree of wettability. The parameter defining the wettability is the observed contact angle, θ . The lower the contact angle, the higher the wettability. This angle is measured between the vertical tangent from the drop edge and the substrate surface, as shown in Figure 10.2. The angle is always measured on the drop side of the tangent.

As will be described later, many contact angle analysis techniques (e.g., half-angle and Bikerman's method) depend upon the spherical nature of the sessile drop. With this as a foundational requirement for the validity of the analyses, it is important to determine if the assumption of a spherical drop shape is valid. Extrand and Moon⁶ have analyzed the boundaries of this spherical domain and they report an expression for the maximum volume of a sessile drop above which gravity begins to distort the spherical drop shape. Equation (10.2) gives the maximum spherical volume for a sessile drop of a given density and surface tension:

$$V_{\max} = \frac{\pi}{48} \left(\frac{\gamma_L}{\rho g} \right)^{3/2} \tan(\theta/2) \left(3 + (\tan(\theta/2))^2 \right) \left[\left(1 + 8 \frac{(\sin \theta)^2}{1 - \cos \theta} \right)^{1/2} - 1 \right]^3 \quad (10.2)$$

Equation (10.2) is plotted in Figure 10.3 to show the maximum spherical volume ($V_{\max}$) of a sessile water drop as a function of contact angle. One can see

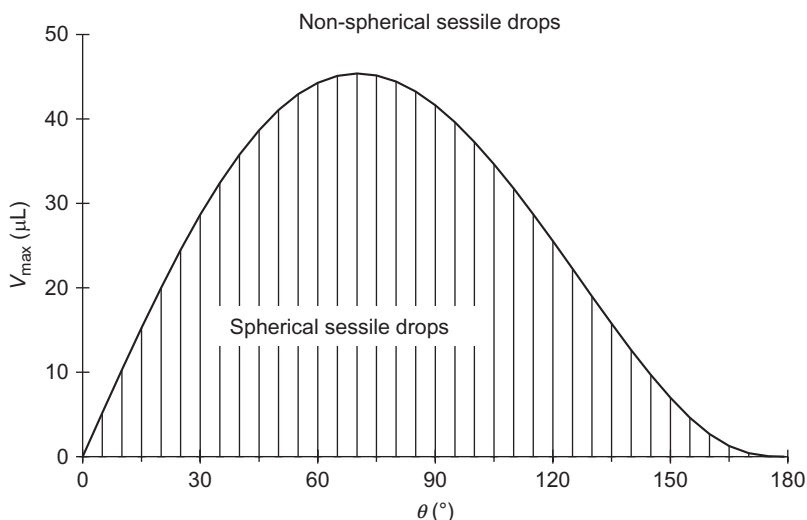


FIGURE 10.3 Maximum spherical volume vs. contact angle for water. Volumes above the curve deviate from sphericity due to gravity.

that for water drops of 10 μL the spherical domain covers drops expressing contact angles from 10° to 140°. The same range of contact angles is valid for very small 1- μL drops of diiodomethane, which illustrates the impact of the lower surface tension and larger density of this liquid.

For a smooth, planar surface, the contact angle value depends on the liquid surface tension, the surface free energy of the solid, and the interactions between the two materials. The equilibrium value of the contact angle, also known as the wetting angle, satisfies Young's equation,⁷ which may be written as:

$$\gamma_S = \gamma_{SL} + \gamma_L \cos \theta \quad (10.3)$$

where γ_S is the surface free energy of the solid, γ_{SL} is the interfacial free energy between the solid and the liquid, and γ_L is the liquid surface tension, as shown in [Figure 10.2](#). Depending on the value of θ , different wetting regimes can be considered as follows:

- $\theta = 0^\circ$: complete or full wetting
- $0^\circ < \theta \leq 90^\circ$: partial wetting
- $90^\circ < \theta < 180^\circ$: partial dewetting
- $\theta = 180^\circ$: complete or full dewetting.

The case of partial wetting is most frequently encountered. The case of complete wetting is generally found for a low surface tension liquid on a solid surface, or for the majority of liquids on clean inorganic surfaces. The case of partial dewetting is encountered with specific, and frequently engineered,

systems, while the case of complete dewetting is only a theoretical situation that may not happen in practice. In order to predict the behavior of a liquid in contact with a solid surface, the free energy of the dry solid must be compared with the free energy of the wetted surface. The difference between these two energies is called the spreading parameter, S .⁸ Its value can be expressed in terms of the parameters defined in Young's equation:

$$S = \gamma_S - (\gamma_{SL} + \gamma_L) \quad (10.4)$$

If $S \geq 0$, the liquid fully wets the solid and the observed static contact angle is zero. When $S < 0$, the liquid does not fully wet the solid and the observed static contact angle achieves a finite value.

The above definitions apply to the solid/liquid/vapor contact angle. A similar set of definitions can be applied to describe a solid/liquid/liquid contact angle, where two immiscible liquids wet a solid substrate. In this case the liquid surface tension is replaced by the interfacial tension between the two liquids and the surface free energy of the solid is replaced by its interfacial free energy when wetted by one of the liquids. Care must be taken to specify the extent of mutual saturation of the two liquids. This is the equivalent to considering the vapor phase saturation for a solid/liquid/vapor system.

According to Equation (10.3), the contact angle of a liquid on a surface is exactly defined by the interfacial free energies of the solid/vapor, liquid/vapor, and solid/liquid interfaces. This ideal case has not been observed: wetting angles for a specific system are observed to adopt a continuum of values between two extreme limits. These limits are referred to as the advancing contact angle, θ_A , observed when the stationary liquid has just started to advance across the solid surface, and the receding contact angle, θ_R , observed when the stationary liquid has just started to recede across the solid surface. These limiting values are considered to be reproducible for a given solid/liquid (and vapor) combination, or for a solid/liquid/liquid combination.

Several hypotheses have been formulated to explain the wetting hysteresis ($\theta_A - \theta_R$), whose sources are considered to be:

- surface roughness on the solid surface⁹;
- chemical heterogeneities on the solid surface (can be generated by adsorption of contaminants¹⁰);
- rearrangement of the solid surface when in contact with a liquid (e.g., a polar polymer in contact with a polar liquid)^{11,12}; and
- local deformation of the solid surface.^{13–15}

The no wetting situation with a finite observed contact angle is completely free of wetting hysteresis. The case of complete wetting is, by definition ($\theta = \theta_A = \theta_R = 0$), free of hysteresis. Low wetting hysteresis ($\theta \approx \theta_A \approx \theta_R$) has been observed for some highly uniform solid surfaces with ultra-low surface free energy, such as a silicon wafer coated with self-assembled monolayers (SAMs) of a hydrophobic silane.¹⁶

Rodriguez-Valverde et al.¹⁷ have tried to define the most stable contact angle (MSCA) which is achieved by mechanical vibration of the surface with a sessile drop on it. Their contact angles after vibration were reproducible with standard deviations $\leq 2^\circ$.

Surfaces are classified into two types: high energy surfaces and low energy surfaces. High energy surfaces include metals, metal oxides, glasses, and ceramics. Low energy surfaces are primarily organic and polymeric materials. Liquids generally spread on high energy solid surfaces, since the liquid surface tension is considerably less than the surface free energy of the solid substrate.^{18,19}

Some exceptions to this rule exist. Gold and other metal surfaces absolutely free of oxides, as well as HF-etched silicon, are not completely wetted by water. The issue of the wettability of gold by water has been a controversial one, with both wettability and non-wettability of gold being reported in the literature.²⁰ For a gold surface, standard cleaning procedures to remove the residual organic contamination, such as oxidation, did not alter the hydrophobic nature of the gold surface. Heating the gold surface to 1000 °C in air also left it hydrophobic. At the time, the explanation was made that organic materials were strongly adsorbed onto gold and were therefore more difficult to remove. However, this was not consistent with the generally low adsorptive character of a gold surface. During an investigation of gold wetting, it was found that the gold could be made wettable by anodically growing an oxide layer on its surface.²¹ When heated, the unstable oxide decomposed to gold and oxygen and the surface reverted to its hydrophobic behavior. Thus, an oxide-free gold surface was non-wetting to water. The formation of an oxide rendered the surface hydrophilic. A similar behavior is found for a silicon surface, which is hydrophobic after the oxide has been removed and becomes hydrophilic when the surface is re-oxidized.

The wettability behavior of metals with water was confirmed by Erb.²² Continual condensation of water onto metal surfaces for many thousands of hours showed that only noble metals continued to have dropwise condensation of water, indicating hydrophobic behavior. Indeed, aluminum, chromium, and other metals known to form oxides had hydrophilic surfaces after prolonged exposure to the water vapors. Fowkes developed a theoretical confirmation of the inherent hydrophobic nature of oxide-free metal surfaces by considering the dispersion forces involved in wetting of solid surfaces.^{23,24} However, the controversy is not fully resolved, since Schrader²⁵ has reported that a pure gold surface prepared in ultra-high vacuum is wettable by water.

Any test of surface cleanliness involving wettability by water cannot be used on metal surfaces that have an unknown oxide layer. It is tempting to assume that any clean metal oxide surface would be hydrophilic, but even this rule may have some exceptions.

Extensive contact angle data are available in the literature. Most measurements are of advancing contact angles. Generally, small advancing contact angles on metal surfaces are obtained only on extremely clean surfaces.

10.4 INSTRUMENTATION

The most common method of contact angle measurement is the direct observation of a sessile drop. The contact angle is determined from the profile of the drop where it meets the solid surface. This may be done directly with a contact angle goniometer equipped with an eyepiece. The drop is viewed from the side and a tangent is aligned with the liquid surface at the point where it meets the solid substrate. Alternatively, a drop of known volume is placed on the solid substrate and the drop is viewed from above. The diameter of the drop gives its contact angle. A non-circular drop profile indicates a non-uniform wettability on the solid substrate. The advancing and receding contact angles can be measured by adding or withdrawing liquid from the drop. Neither of these techniques requires prior knowledge of the surface tension or the density of the liquid. The accuracy of both methods is typically $\pm 1^\circ$, which is also the typical error bar in contact angle measurements on a variety of uniform surfaces.

Contact angle measurements provide a simple, rapid, and non-destructive technique for probing the surface chemistry of a solid surface.²⁶ The wettability is very sensitive to the exposed surface chemistry. Wetting properties have a high degree of correlation to surface reactivity and adhesion properties of the solid substrate.²⁷

The heart of a contact angle goniometer is an optical bench. Some examples of contact angle goniometers are shown in [Figure 10.4](#). At one end of the bench is the light source and at the other end we find the measurement optics, consisting of a binocular with a cross-hair and a goniometer in the eyepiece. In between these two is placed the solid substrate and a syringe to dispense the probe liquid. The solid substrate is maintained horizontal for sessile drop measurements and may be inclined perpendicular to the optical axis for advancing and receding contact angle measurements. The basic version of the instrument is equipped with a micrometer-driven syringe and a mechanical stage, allowing an accurate dosing and reliable positioning of the liquid drop on the solid substrate. The eyepiece contains a protractor scale with 1° division for easy measurements. This flexible design allows many different accessories to be adapted. Manual measurements can be made with the goniometer eyepiece, which a trained operator can use to determine contact angles rapidly and easily. Manual contact angle measurement is rapid and allows direct observation of the liquid drop behavior on the sample. This technique was patented where the eyepiece was replaced with a graduated projection screen.²⁸

To achieve automation, the eyepiece in the goniometer bench may be replaced by an image acquisition system connected to a computer, allowing rapid sample analysis and documentation. Image processing software is used to analyze the acquired images and reliably define the drop boundary. The software can analyze not only images of sessile drops, but also images of enclosed bubbles and pendant drops. It can calculate several parameters of interest from these drop shapes, such as static or equilibrium contact angles, advancing and

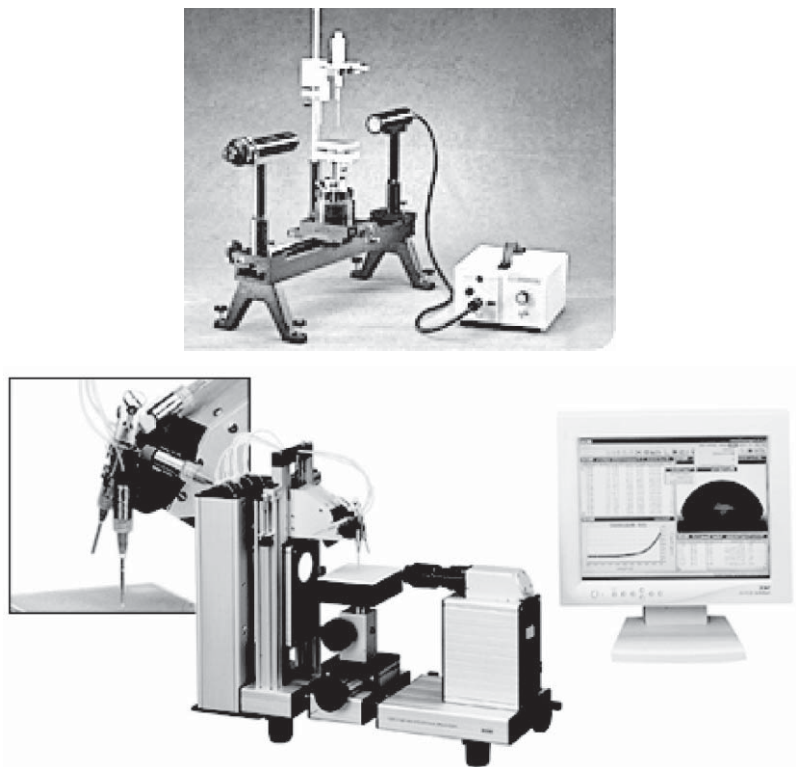


FIGURE 10.4 Examples of contact angle goniometers.

receding contact angles, liquid surface tension, interfacial tension between two liquids, surface free energy from contact angles, and Zisman critical wetting tension¹⁹ from the contact angles of a homologous series of liquids, such as alkanes.

Neumann and coworkers²⁹ are active in writing routines to analyze images obtained by commercial goniometers to increase the accuracy of contact angle measurements to better than 0.2° . Their numerical approach is interesting in that it allows for a syringe to protrude from the top of the drop. This enables the addition and removal of liquid so that the advancing and receding contact angles can be measured.

Laboratory-based goniometers suffer from two drawbacks: (1) high cost and (2) they are in the laboratory. Durkee and Kuhn³⁰ reviewed methods that are relevant to the shop floor. These include the dyne pens, the water-break test, the Nordtest poly 176 method, and the Bikerman method. The authors draw on their extensive background in industry to report cases where each of these tests were used. They offer a well-deserved criticism of the water-break test while giving ample reason to move toward other inexpensive tests that are more meaningful with respect to cleanliness evaluation such as the Bikerman method.

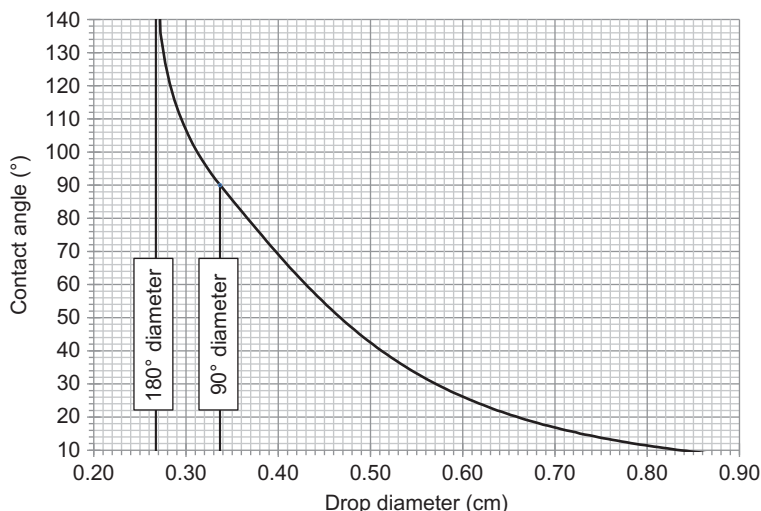


FIGURE 10.5 Nomogram of the Bikerman equation for a 10- μ L sessile drop, which allows the conversion of drop diameter directly to contact angle. The diameter of the drop at 180° contact angle and 90° contact angle as seen from above are indicated on the nomogram.

The Bikerman method requires an accurate delivery of a volume of liquid and an accurate measurement of the contact diameter of the sessile drop. Bikerman³¹ used talc to leave a ring after the drop evaporated, but this is unacceptable in many cases. Miller³² used a calibrated magnifying reticle and a nomogram similar to that shown in Figure 10.5, where the diameter of the drop and the volume of the drop are used to read the contact angle from the chart.

The Bikerman method was also used on nano-sized droplets by Dillingham and coworkers³³ to measure siloxane mold-release contamination from peel ply laminates prior to adhesive bonding. The practical aspects of the Bikerman method for measuring wettability are extolled by Dillingham, in contrast to the dynamic contact angle (DCA) or laboratory-based goniometer methods, which were unsuitable for field use.

Bikerman's equation is valid only for drops expressing contact angles up to 90°. Others^{34,35} have derived the equation for the >90° case, so that there are no theoretical issues with using the Bikerman method for non-wetting drops. In practice, however, the drop diameter measurement for non-wetting drops leads to unacceptable uncertainty in contact angle. This can be seen in the slope of the nomogram line as the drop diameter approaches the 180° drop diameter in Figure 10.5.

Recent advancements in digital imaging, computing, and image analysis software have made it possible to construct home-built goniometers with acceptable precision. Inexpensive USB (Universal Serial Bus) microscopes and LED (light-emitting diode) light sources are readily available. Williams et al.³⁶ describe the

use of a freely available image analysis software package (ImageJ³⁷) that has two plug-ins for use in analyzing side-view images of sessile drops. The plug-ins allow the user to select the Low Bond Axisymmetric Drop Shape Analysis model,³⁸ the spline-fitting Drop Snake model,³⁹ or the ellipse and circle fitting technique.⁴⁰ The half-angle technique can also be used without any special software package by counting the pixels across the base of the drop, d_{base} , and from the surface to the apex, h , of the drop when viewed in profile (Figure 10.2):

$$\theta = 2 \arctan \left(\frac{2h}{d_{\text{base}}} \right) \quad (10.5)$$

It is difficult to get a profile view of a sessile drop on large surfaces without some additional optics. Prisms have been used to enable side views of sessile drops, and the contact angle was measured with a microscope containing an eyepiece with a protractor.⁴¹ The use of a prism vs. an inexpensive bent metal mirror has been evaluated with USB microscopes.³⁴ And the usefulness of cell phone cameras with the Bikerman method has also been reported with positive results.⁴² Two provisional US patents have recently been filed which turn smart-phone devices into contact angle measurement platforms—one using the Bikerman method⁴³ and the other using the Langmuir method.⁴⁴

The Langmuir method was first reported by Irving Langmuir and Vincent Schaefer in 1937.⁴⁵ It has since been patented in various forms⁴⁶ and marketed as the Contact- θ -Meter.⁴⁷ The Langmuir technique measures the minimum angle of the light reflected from the front edge of a sessile drop. This minimum reflection angle will define the normal to the drop surface at the base contact point. This normal is 90° to the tangent that defines the contact angle. Therefore, the contact angle is 90° minus one-half of the minimum reflection angle.

The wetting or contact angle can also be determined indirectly from capillary rise measurements, such as the Wilhelmy plate technique, where a plate is immersed vertically into the liquid until it partially covers the plate. The liquid forms a meniscus, whose shape is determined by Laplace's equation, as shown in Figure 10.6.

From this equation, a relationship can be derived for the contact angle as a function of the capillary rise height, c , of the liquid on the plate:

$$\sin \theta = 1 - \frac{\rho g}{2\gamma_L} c^2 \quad (10.6)$$

where ρ is liquid density and g the acceleration of gravity. This principle has been used to develop DCA analyzers (DCAAs), which measure both advancing and receding contact angles on solids. Instead of measuring the capillary rise, a DCAA uses a microbalance to measure the wetting force exerted by the liquid

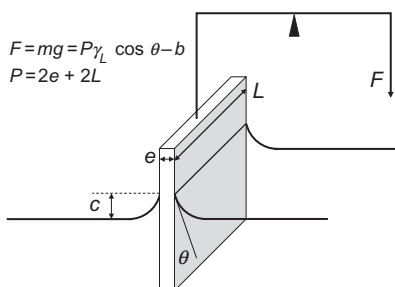


FIGURE 10.6 Principle of contact angle measurement with the Wilhelmy plate.

meniscus on the solid surface. This force corresponds to the weight of the liquid meniscus, w , which satisfies the following equation:

$$w = P\gamma_L \cos \theta \quad (10.7)$$

where P is the solid perimeter. The solid plate hangs from the microbalance, which measures the vertical force, F , acting on the plate. The substrate is generally small and light, allowing optimal functioning of the microbalance. The contact angle is derived by measuring the vertical force on the sample during immersion and emersion of the sample from the liquid medium. The measured force, F , is composed of the sample weight, mg , where m is the sample mass, the buoyancy force, b , from the displaced liquid, and the meniscus force, $P\gamma \cos \theta$, as shown in Figure 10.6. Thus, the force, F , measured by the microbalance is

$$F = mg + P\gamma_L \cos \theta - b \quad (10.8)$$

The programming of a DCAA allows either to determine the liquid surface tension when the wetting angle is zero, or to determine the advancing and receding wetting angles with a liquid probe of known surface tension. The advancing and receding contact angles correspond to the immersion or emersion of the solid from the liquid respectively. The calculation performed by a DCAA takes into account the buoyancy correction, b , generated by the solid substrate displacing the bulk liquid. An advantage of a DCAA is that wetting measurements can be performed on flat as well as on cylindrical solids. Further, since the heart of the equipment is a microbalance with a precision of 10^{-8} - 10^{-9} N, the wetting properties of extremely thin fibers, such as glass fibers or hair, can easily be measured.

The observed contact angle on a solid substrate depends not only on the intrinsic surface characteristics of the substrate, but also on any material adsorbed onto the substrate surface. Both the advancing and receding water contact angles on freshly cleaned glass or platinum surfaces are 0° . The recommended cleaning procedure for these materials is exposure to the hot portion of a gas flame. Following cleaning, exposure of the clean plate to ambient atmosphere generally results in a rapid increase in advancing contact angle. This

increase is caused by a lowering of the surface free energy as organic contaminants contained in the atmosphere are adsorbed onto the clean surfaces.

The DCA analysis is well suited to making contact angle measurements on solid surfaces: the contact angle is averaged over the line formed by the meniscus and the measurement is recorded during immersion of the entire substrate. This allows a rapid examination of the wettability of the entire surface. Surface inhomogeneities are easily detected from distortions of the contact line, which generate corresponding fluctuations in the force measured by the microbalance. Moving the solid downward and upward measures advancing and receding contact angles respectively at a given displacement speed of the contact line. A zero receding contact angle on the solid substrate is easily determined. When this occurs, the force generated by the meniscus is equal to the product of the solid perimeter and the liquid surface tension.

The DCAA (Figure 10.7) uses the microbalance technique coupled with software to provide a simple and flexible analytical tool. Its measurements are based on a highly precise and accurate microbalance that provides high sensitivity and reproducibility. The force resolution is 10^{-9} N (corresponding to 10^{-7} g), allowing the study of wetting on single fibers. For thin and lightweight solid substrates, the apparatus can measure advancing and receding contact angles, liquid surface tension, interfacial tension between two immiscible

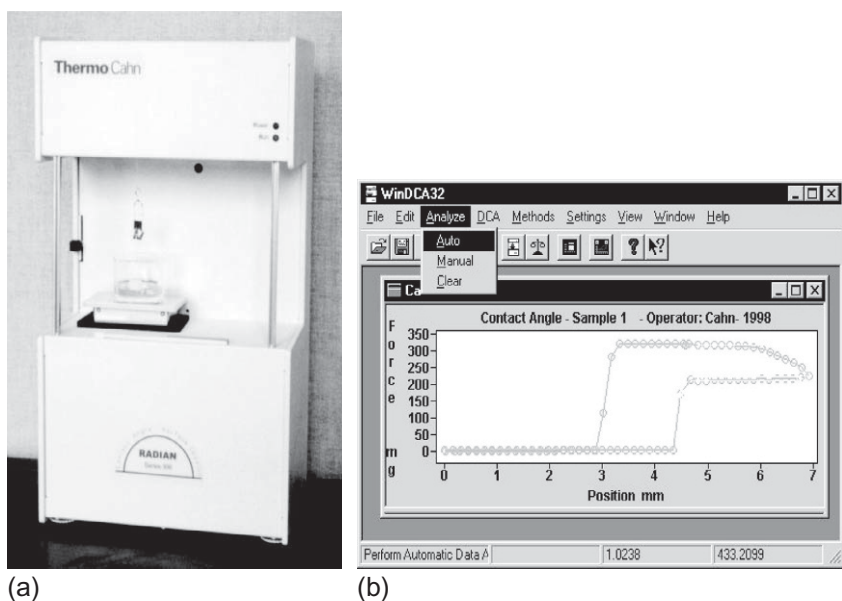


FIGURE 10.7 Example of a DCA analyzer used for surface tension and dynamic contact angle measurements. This system can be applied to many types and geometries of solid surfaces, including single fibers. (a) Shows the basic equipment (microbalance) and (b) shows the force diagram in advancing and receding modes leading to the contact angle hysteresis.

liquids, surface free energy from contact angles, and Zisman critical wetting tension. It can also be used to measure wicking rate and contact angles on granulated materials.

10.5 EXAMPLES OF APPLICATIONS

Contact angle measurements can be used to investigate the result of a cleaning process, contributing to the optimization of cleaning procedures. Measuring both advancing and receding contact angles allows a range of adsorption and contamination processes to be monitored. This can avoid the use of more laborious and occasionally less informative methods of analysis, such as ultra-high-vacuum analysis techniques, which can eventually be used as a final method of confirmation.

High surface free energy materials, such as metals, glass, and ceramics, when exposed to ambient air are generally coated with a partial layer of organic molecules. These coatings reduce the surface free energy of these materials from their intrinsic values and render the final surfaces hydrophobic. Such contamination may generate large advancing contact angles, up to about 80° . Degreasing the surfaces generally reduces the advancing contact angle to about 30° , which corresponds to residual adsorbed hydrocarbons, most of which generally come from the degreasing agent. To obtain a higher wettability, all residual adsorbed organic molecules must be removed. Only highly clean oxidized metal surfaces are fully wetted by water, with both the advancing and receding contact angles being 0° . Small levels of hydrophobic contamination on a hydrophilic surface lead to a rapid increase in the observed advancing contact angle.⁴⁸ For example, a full coverage of a planar surface by small organic molecules can be achieved with only 1 g per 1000 m^2 of surface. Even less than full-layer coverage is sufficient to cause a finite advancing contact angle on the contaminated surface.

In industrial environments, metal surfaces are generally contaminated, making their advancing contact angles difficult to predict. In most cases, these fall in the range from 40° to 80° . The receding contact angle is generally smaller than 20° , thus generating a large contact angle hysteresis. Hence, both advancing and receding contact angles are needed to completely describe the wettability of these surfaces.

High-energy solid surfaces such as metallic oxides or metals are highly prone to contamination from organic vapors in the atmosphere. The large change in the surface wettability toward water makes the presence of contamination easy to detect.

Polymeric surfaces are typically low-energy, hydrophobic, and difficult to coat or bond. For this reason, plasma treatment is often used to increase the surface free energy of polymeric materials. Dillingham and coworkers⁴⁹ have shown that water continues to be an excellent probe liquid for measuring the

surface energy of plasma-treated polymeric surfaces, and can predict if the coating or adhesive will stick to the polymeric surface.

O'Kane and Mittal⁵⁰ have used surface wettability to study hydrophobic organic contamination and the rate of contaminant adsorption by rhodium and iron–cobalt (Fe–Co) alloy surfaces. Auger electron spectroscopy provided identification of the impurity elements and an estimate of their concentrations on the metal surfaces. Figure 10.8 shows the rate of adsorption of hydrophobic contaminants on the surface of Fe–Co as measured by the water contact angle. A freshly evaporated surface of Fe–Co shows a slow rate of adsorption which is indicated by the increase of the contact angle from less than 5° to 51° over a period of 15 days. A badly contaminated surface of Fe–Co (initial water contact angle of 83°) was plasma cleaned to produce a contact angle of less than 5°. After exposure to laboratory air, the contact angle increased to 50° in 9 days (Figure 10.8). This apparent higher rate of adsorption of impurities for the plasma-cleaned surfaces may be caused by active surface sites generated.

The change in surface wettability for various surface cleaning conditions is reported in Table 10.1.

The results of Table 10.1 show that cleaning with organic solvents was not as effective as plasma cleaning in the removal of surface contaminants. The results concerning the rhodium substrate showed that the surface wettability was dependent on the length of time in the plasma. Plasma-cleaned Fe–Co surfaces were equivalent to freshly evaporated surfaces in terms of water wettability (contact angle lower than 5°, Figure 10.8). No significant difference was

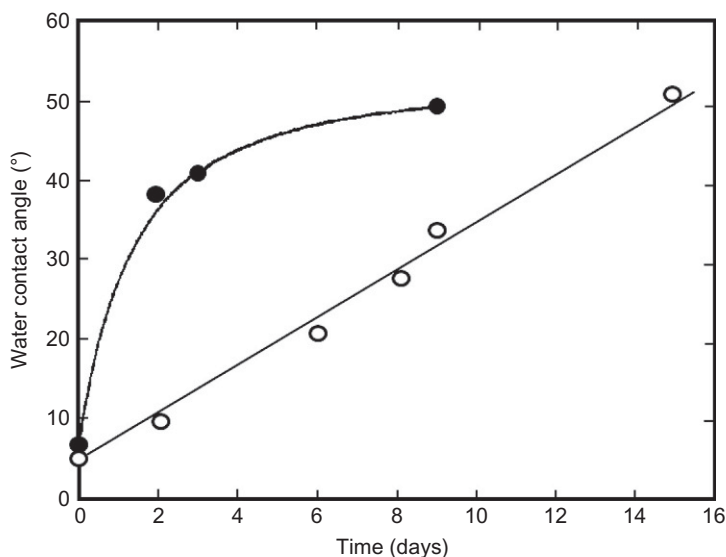


FIGURE 10.8 Rate of adsorption of hydrophobic contaminants on Fe–Co alloy surfaces. Filled circles: plasma-cleaned surface; open circles: freshly evaporated surface.⁵⁰



LIVE GRAPH
Click here to view

TABLE 10.1 Change in Water Wettability as a Function of Surface Cleaning Conditions⁵⁰

Type of Surface	Method of Cleaning	Water Contact Angle (°)
Fe–Co	No cleaning of contaminated surface	96
	Organic solvents	80
	Organic solvents and helium–oxygen plasma (10 W for 1 min)	<5
	Freshly evaporated Fe–Co surface	<5
Rhodium	Vapor degreasing	38
	Vapor degreasing+ argon plasma (40 W for 1 min)	19
	Vapor degreasing+ argon plasma (40 W for 3 min)	13
	Vapor degreasing+ argon plasma (40 W for 6 min)	8

observed in the cleaning efficiency of the plasma when argon was substituted for the helium–oxygen mixture. Argon was preferred to avoid the possibility of metal oxidation in the plasma.

The contamination rates for six metallic oxides were probed by Matsunaga and coworkers.^{51,52} The solids examined were: crystalline quartz glass (SiO₂), titanium oxide (also known as rutile, TiO₂), chromium oxide (also known as chromia, or Cr₂O₃), alumina (Al₂O₃), nickel oxide (NiO), and strontium titanate (SrTiO₃). The samples were polished with an emery paper, followed by polishing with diamond powder. They were then cleaned with ultrasonic agitation in a detergent solution, followed by thorough rinsing in double-distilled water. Following cleaning, the metal oxide surfaces were fully wettable by water. These samples were then dried for 1 h in an enclosure containing P₂O₅. To generate surface contamination by adsorption of organic molecules, the clean samples were exposed to liquid-paraffin vapor. This was done in a Petri dish with two small containers. One container was filled with paraffin and the second container was filled with water to generate a humid atmosphere. Figure 10.9 shows the water contact angles plotted against time for clean samples exposed to paraffin vapor in a dry atmosphere and Figure 10.10 shows data from clean samples exposed to paraffin vapor in an atmosphere saturated with water vapor. In a dry atmosphere, the most rapidly contaminated sample was rutile. The contact angles on silica and alumina surfaces increased more slowly. In the wet

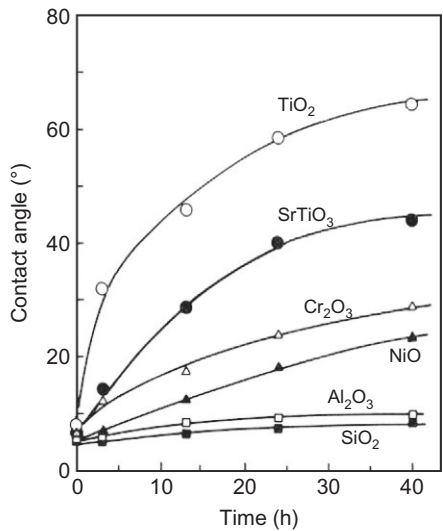


FIGURE 10.9 Contamination rates (in terms of increase in water contact angle with time) on metallic oxide surfaces under dry atmosphere.⁵²

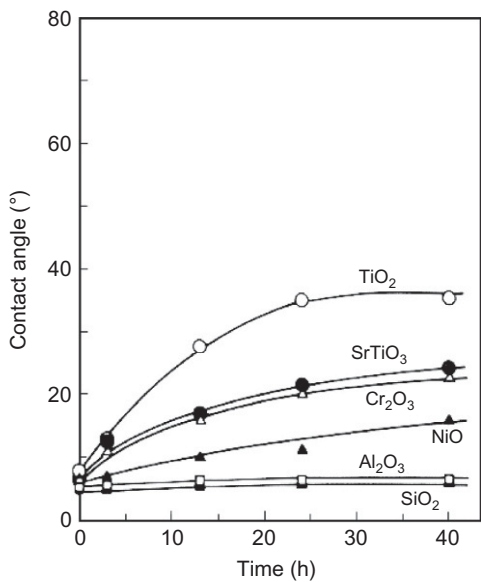


FIGURE 10.10 Contamination rates (in terms of increase in water contact angle with time) on metallic oxide surfaces under wet atmosphere.⁵²

atmosphere, the extent of contamination was significantly reduced for all the surfaces, with the largest attenuation recorded for the silica and alumina surfaces, with only minor increase in their contact angles after 40 h of exposure to paraffin vapor. However, the trends in the contamination rate versus material were the same for both dry and humid atmospheres. We may deduce from this study that a surface having a larger affinity to water will be contaminated more slowly by adsorption of organic molecules. A similar conclusion was found following a study of the wettability of glasses with different compositions. This study is described in [Section 10.6](#).

It is tempting to deduce that all clean oxide surfaces are hydrophilic. However, some caution is necessary. White⁵³ cleaned a polished fused silica surface by degreasing and boiling in 30% hydrogen peroxide. He then heated the cleaned surface in oxygen at various temperatures. By measuring the contact angle of water immediately after cooling to room temperature, he showed that wettability of the surface decreased with increasing heating temperature, as shown in [Figure 10.11](#). Cleaning the silica surface by etching with NH_4F – HF solution generates even lower surface wettability than after heating in oxygen. This increase in the water contact angle with heating is probably caused by dehydration of the silica surface, forming siloxane bridges (Si-O-Si) from the neighboring silanol (Si-OH) sites. The siloxane groups have a more hydrophobic behavior than the original silanol groups (Si-OH). The higher contact angle values measured following fluoride etching are believed to be caused by the formation of silicon fluoride (Si-F) bonds. [Sonders et al.](#)⁵⁴ give an interesting example of the effect on the wetting properties of ion exchange at the surface of soda-lime glass, which has been cleaned in sulfuric acid containing potassium dichromate. Adding a trace of sodium fluoride to the sulfuric acid causes the OH^- groups to be replaced by F^- ions at the glass surface. After a

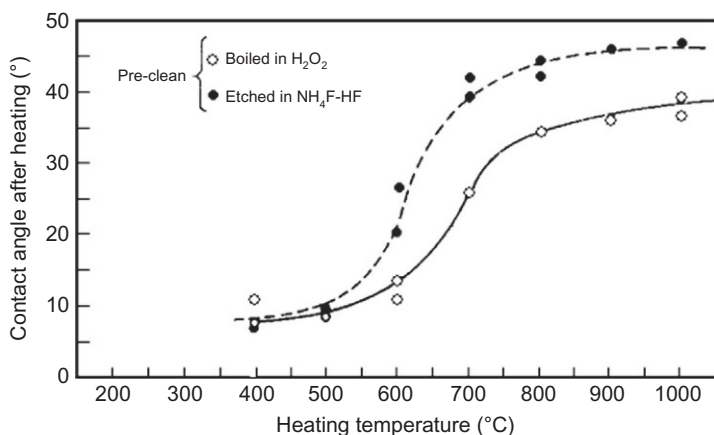


FIGURE 10.11 Effect of heating in dry oxygen on water contact angles on fused silica plate.⁵³

few seconds fluoride ions have replaced the hydroxyl groups and the acid forms little droplets on glass similar to water on a greasy surface. As for the fluoride-etched silicon wafer, the Si-F groups are hydrophobic, leading to higher water contact angles on the silica surface.

As seen above, not all clean metal oxide surfaces are wettable by water. Some caution must be used when interpreting contact angles on these surfaces. However, it has been found that all clean metal oxide surfaces are fully wettable by water, provided they are not heated to high temperatures.

The wettability of oxide surfaces is a very sensitive indicator of the presence or absence of organic contamination on the surface. If we expose clean oxide surfaces to ambient air containing organic contaminants, we expect the contact angles to increase with time. A study was made of the contamination rate on a number of different surfaces.⁵³ Metal surfaces were prepared by polishing high-purity metal samples to a mirror finish with alumina powder, followed by degreasing in trichloroethylene and acetone. The samples were then heated in air at 400–500 °C for 30 min, completely removing any residual organic contaminants and growing an oxide layer on the metal surface. Glass substrates were cleaned with dichromate–sulfuric acid cleaning solution and then fired in air in the same manner as the metals. A clean mica surface was obtained by cleaving a piece of mica immediately before using. The contact angle of deionized water, as measured by the reflection goniometry technique, was between 3° and 8° on all surfaces immediately after cleaning. These surfaces were then exposed to laboratory air for various periods of time and the contact angle of water measured after each exposure. Figure 10.12 shows the water contact angles as a function of exposure time to laboratory air.⁵³ The contact angles on metal surfaces, particularly

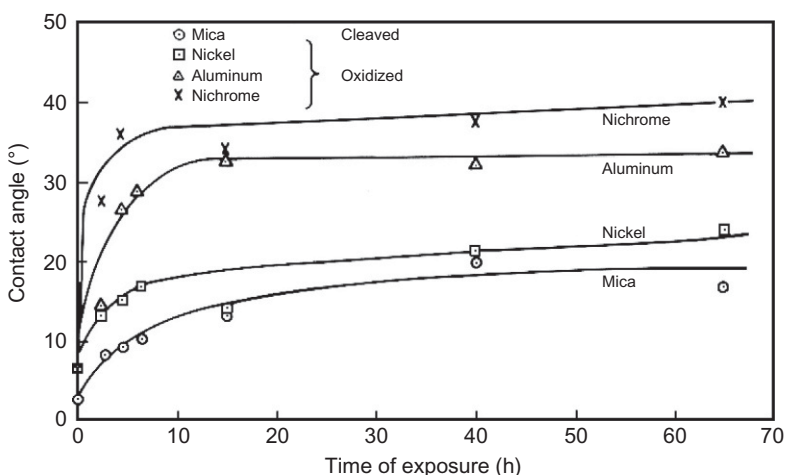


FIGURE 10.12 Contamination rates (in terms of increase of water contact angle with time) of surfaces exposed to laboratory air.⁵³

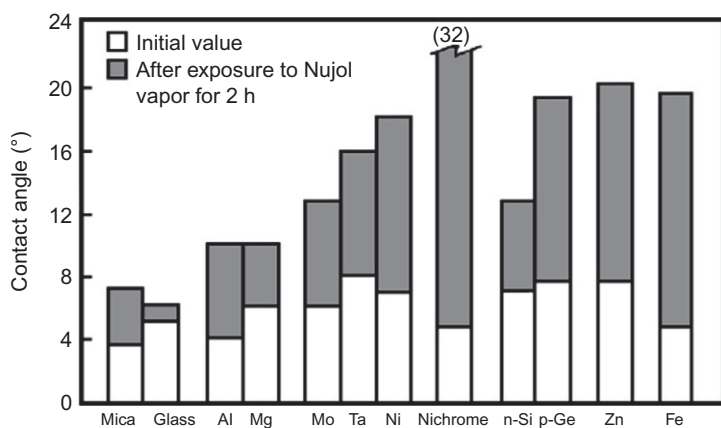


FIGURE 10.13 Water contact angles on oxidized surfaces exposed to Nujol.⁵³

nichrome (a nickel and chromium alloy) and aluminum, attain higher final values than those on mica. They also show a higher initial rate of increase, suggesting that the surfaces are not being contaminated at the same rate.

A consistent contaminating atmosphere was used for an experiment designed to avoid the contaminant variations in ambient air. A number of surfaces were cleaned by White,⁵³ using the processes described above. The samples were then exposed to Nujol (a mineral oil) vapor for 2 h in a closed container. The measured values of water contact angles are shown in Figure 10.13. Contact angles on glass and mica show only a small increase, while aluminum and magnesium show higher values. Transition metals show the largest increase in measured contact angles. The measured water contact angles grow quickly with increasing level of organic contamination up to a saturated value (think Langmuir isotherm), indicating the various surfaces react differently to the available organic contamination in the atmosphere. This confirms the adsorption phenomenon of organic contamination, where each metal oxide has its specific rate and final coverage of adsorbed contamination. From the lower increase in observed water contact angles, mica and glass show lower levels of adsorbed contamination when compared to the transition metal oxides. From this study, we may infer that the transition metal oxides are the most sensitive for detecting organic contamination, since they show the greatest change in wettability when exposed to organic vapors. This means that a clean oxidized nichrome surface with its very reactive transition metal oxides will give a better indication of ambient organic contamination than less adsorptive surfaces featuring main group oxides, glass, or mica.⁵³

The concept of organic contamination as an adsorption phenomenon has led to an interesting technique for storing clean surfaces.⁵³ Since oxidized metal surfaces have a higher affinity for organic vapors than glass or mica, they will preferentially adsorb ambient organic contaminants. A container with a

relatively large surface area of a clean metal oxide is expected to maintain an organic-free atmosphere by reason of its high affinity for organic molecules. This was tested by heating degreased aluminum shots to 500 °C in an aluminum desiccator. During this process, organic material would be desorbed from the surface of the aluminum and an aluminum oxide layer would be grown. A piece of mica that had been contaminated by exposure to ambient air was placed into this container and the contact angle measured periodically, probing its level of adsorbed organic contamination. Results showed that the water contact angle on contaminated mica decreased from 15° to 7.5° after 100 h of storage, indicating that organic contaminants were desorbing from the mica surface and adsorbing onto an oxidized aluminum surface. While this cannot be a practical routine technique for reducing contamination on mica or glass surfaces, it is a demonstration of surface adsorption and desorption on different materials.

To examine the effectiveness of various containers for maintaining clean surfaces, samples of nichrome were placed into various containers for 16 h.⁵³ Nichrome was chosen as a test surface, due to its high affinity for adsorbing organic contaminants. Prior to storage, the samples were made fully wettable by water by cleaning and oxidation, as described above. Water contact angles were measured as a function of storage time, probing the contamination level on the surface. Figure 10.14 shows the results obtained for a variety of storage conditions. Exposure to laboratory air for 16 h results in a contact angle of 35°. For the same storage time, the contact angle following storage in a ground-glass stoppered weighing bottle is only slightly lower, indicating only marginal improvement. Storage in an empty aluminum desiccator shows significantly lower surface contamination, as evidenced by the contact angle of only 18°. The storage conditions are further improved by placing activated charcoal or activated alumina inside the desiccator. The best storage condition is obtained when cleaned and oxidized aluminum shots are placed in the aluminum desiccator, where no increase in the contact angle obtained immediately after cleaning is observed. Figure 10.15 shows the effectiveness of storing nichrome and aluminum samples in this environment for 120 h. Water contact angles were measured on the samples at regular intervals and the samples were returned to the storage container after each measurement. It is interesting to note that the contact angle on cleaned nichrome surfaces slowly increases over a period of several days, while the oxidized aluminum surface maintains its original low wetting angle, indicating a lower level of contamination. This result confirms different levels of contamination on different surfaces exposed to the same atmosphere. In a manner similar to water on dehydrating agents, an oxidized aluminum surface eventually becomes saturated with contaminants. The surface must then be regenerated by removing the adsorbed organic contaminants. This can be achieved by degreasing the aluminum container and aluminum shots and heating them in air to 500 °C. For the majority of practical applications, aluminum provides an adequate material for adsorbing ambient organic contaminants and maintaining surface cleanliness.

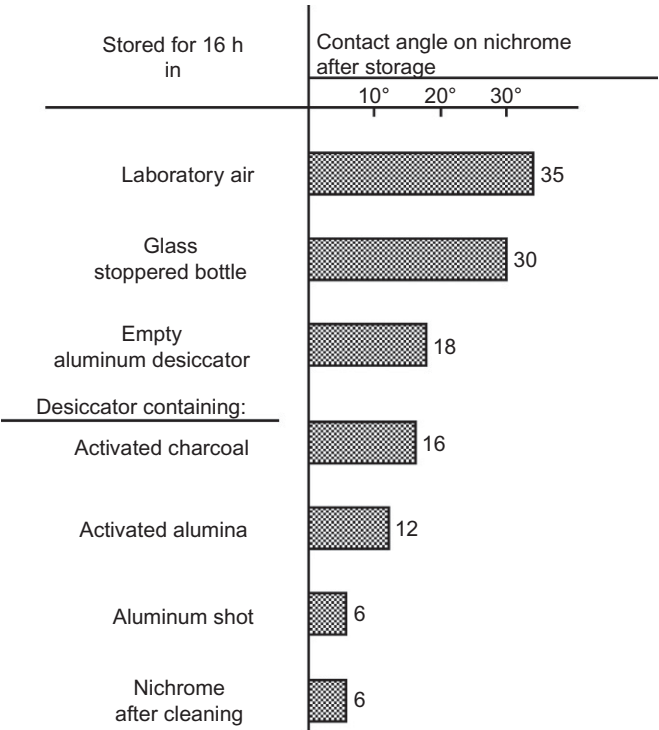


FIGURE 10.14 Evaluation of storage condition on water wettability of nichrome.⁵³

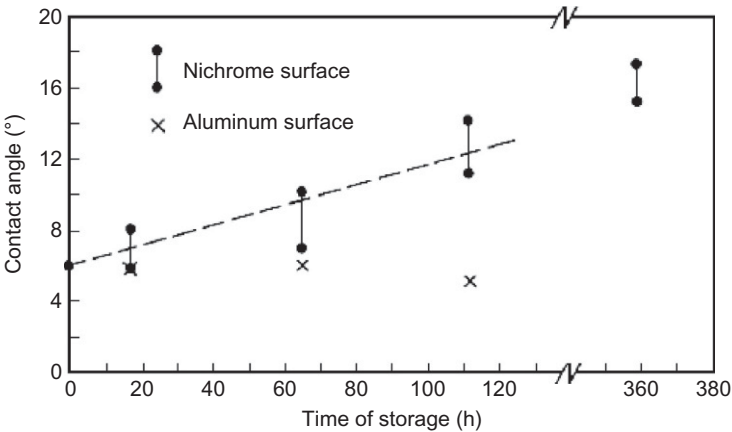


FIGURE 10.15 Water contact angles on surfaces stored in an aluminum container with aluminum as a getter.⁵³

Pohlack and Wendler demonstrated how the initial condition of a glass surface (Schott BK7) influenced its wettability toward water after each step of the cleaning operation.⁵⁵ Glass that had been freshly polished using a pitch polisher, followed by drying on a linen cloth, gave contact angles ranging from 20° to 70°. The two sides of the same sample did not give the same contact angles. When an organic solvent was used to clean the glass, the contact angle decreased but continued to show a relation with the initial value. Glow discharge cleaning always gave low contact angles, as shown in Figure 10.16. Small differences in contact angles, with values near zero, were difficult to detect. Thus, it is not certain whether glow discharge cleaning erases any influence of the initial surface condition. Figure 10.17 shows the increase of the water contact angle measured at different times after glow discharge cleaning.⁵⁵ The cleaning was performed using two different high-voltage power inputs.⁵⁵ The results suggest that more effective cleaning of the glass causes it to be contaminated more rapidly on exposure to a contaminating atmosphere. A saturation level of contamination may be indicated by the asymptotic values of the two curves, suggesting that an improved cleaning leads to a higher level of contamination of the glass surface. The recontamination of clean glass surfaces exposed to a contaminating atmosphere is a fast process. It begins immediately after cleaning and the surface is saturated with contaminant after only a couple of hours.

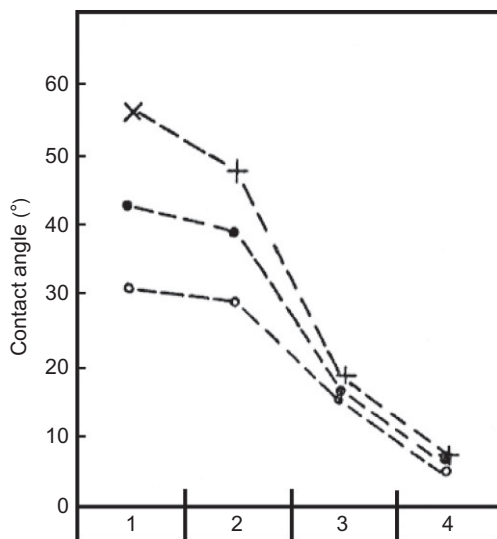


FIGURE 10.16 Change in water contact angle on glass after cleaning as follows: 1, polished and dried; 2, organic solvent; 3, chromic acid; 4, glow discharge.^{55,56} The three symbols correspond to different initial wettability conditions.

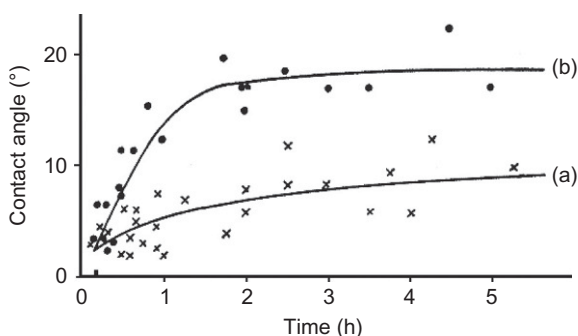


FIGURE 10.17 Change in water contact angle on glass with time after glow discharge cleaning for: (a) 2 kV, 0.2 A, 3.5 min; and (b) 3.8 kV, 0.2 A, 3.5 min.^{55,56}

10.6 RECENT DEVELOPMENTS

The surface contamination of different glass surfaces was studied as a function of glass composition and cleaning process.⁵⁶ The susceptibility of glass surfaces to strongly adsorbing organic contaminants was found to depend both on the glass composition and the cleaning procedure. Silica surfaces show a large difference in their contamination behavior following pyrolysis cleaning or cleaning in a saturated potassium dichromate solution in concentrated sulfuric acid (this solution, known as Chromerge, is prepared by completely dissolving 20 g of potassium dichromate in 90 g of water. To this is slowly added 900 g of concentrated sulfuric acid. The mixing reaction is highly exothermic, exposing the operator to a high risk from spattering. This solution can be reused as long as it remains brown, rather than green, indicating the oxidation state of the dichromate).

In this study, pyrolysis cleaning consisted of heating the glass substrates to 500 °C for 5 h. The pyrolyzed samples were held in a glass rack and placed in a glass container which was closed with folded aluminum foil. Following removal from the oven, the glass samples remained clean for as long as 1 week or longer, provided the lid was not opened. Following opening of the container, the clean glass samples were used within 5 min.

The glass samples were cleaned by immersing them for 20 min in the Chromerge solution at room temperature. They were then rinsed in pure water and dipped for 20 min in a 1:1 concentrated hydrochloric acid/water solution. This was designed to remove chromium ions, which may strongly adsorb organic contaminants from the glass surface. The samples were then rinsed in pure water and dried under a stream of clean nitrogen.

Before cleaning, the glass samples showed no macroscopic organic contamination on their surface. Hence, the samples had only a limited amount of organic contamination, consisting of a monolayer or sub-monolayer coverage of molecules. This layer, removed by the pyrolysis or Chromerge cleaning,

generally has a thickness of the order of 1 nm or less, corresponding to less than 1 g per 1000 m² of planar surface. Following cleaning, glass samples were verified to be fully wettable by water, as a deposited 1- μ l drop of pure water was observed to spread on the sample. On tilting the sample, the water drop migrated easily, leaving a liquid streak behind, which indicated a zero degree receding contact angle.

While aluminoborosilicate Corning code 1737 glass shows some change in its surface properties after pyrolysis or Chromerge cleaning, soda-lime glass shows immunity to contamination after pyrolysis cleaning. However, for the soda-lime glass surface, its contamination behavior is similar to an acid-cleaned silica surface following Chromerge cleaning. Despite the notion of a glass surface being a high-energy substrate that is found to attract ambient organic contamination, these results suggest that the sensitivity to organic contamination depends on the composition of the glass surface. This composition is determined by the bulk glass composition and also by the cleaning procedure. It appears that increasing hydrophilicity of the glass surface may reduce its affinity for adsorbing organic contaminants. This result corroborates the conclusion of the study by Matsunaga and coworkers on the contamination rate of several oxides.^{51,52}

DCA measurements allow measurement of wettability on small cylindrical solids, such as optical fibers with a diameter of 125 μ m. A typical force diagram corresponding to the wetting and dewetting processes is given in Figure 10.18. The wettability of an optical fiber was measured following each step in its

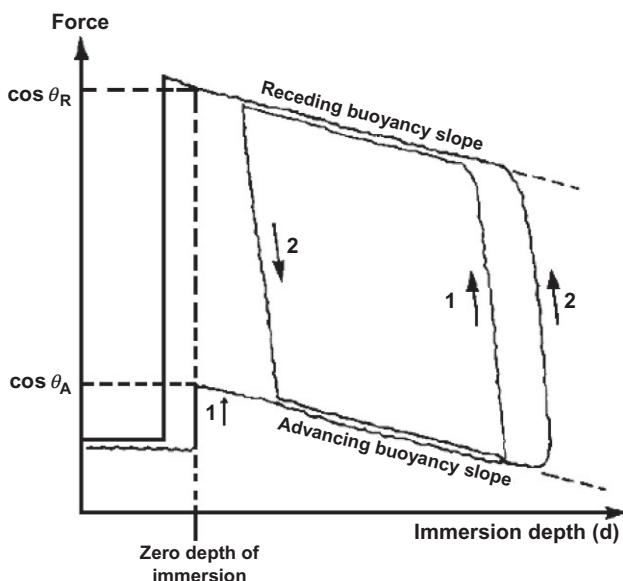


FIGURE 10.18 Measurement of the wetting force with a DCA analyzer during the wetting and dewetting cycles.¹

cleaning, consisting of mechanical and chemical stripping of its cladding. The UV-curable acrylic polymer cladding was first stripped mechanically. The fiber was then cleaned with ethanol and, finally, residual cladding was removed by etching the glass fiber in hydrofluoric acid. The results are presented in [Figures 10.19–10.21](#). The wetting cycle allows determination of the advancing contact angle, while the dewetting cycle gives the receding contact angle. The results indicate an increase of the glass wettability by water and a decrease of the wetting hysteresis, as expressed by the difference between cosines of the advancing and receding contact angles.

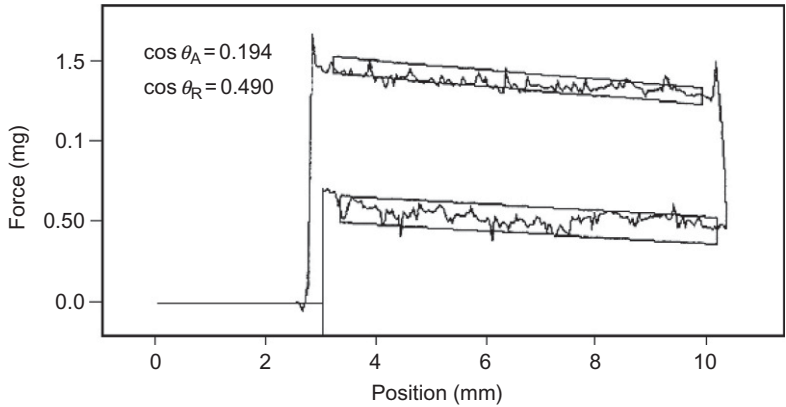


FIGURE 10.19 Wetting and dewetting cycle of water on an optical fiber of 125 μm diameter after mechanical stripping ($\theta_A = 78.8^\circ$, $\theta_R = 60.4^\circ$).¹

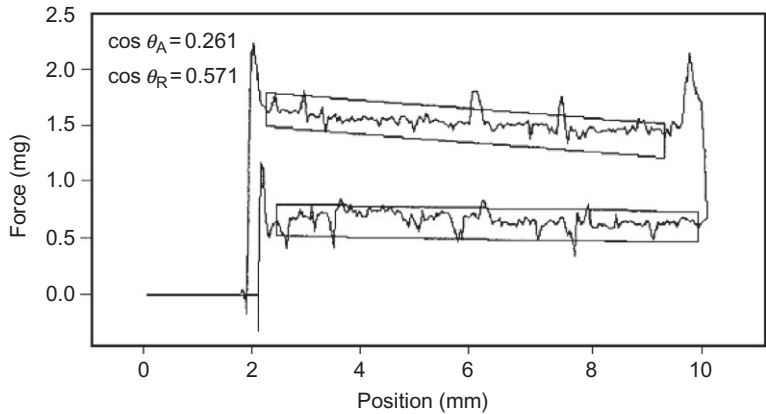


FIGURE 10.20 Wetting and dewetting cycle of water on an optical fiber of 125 μm diameter after mechanical stripping and ethanol washing ($\theta_A = 74.9^\circ$, $\theta_R = 55.2^\circ$).¹

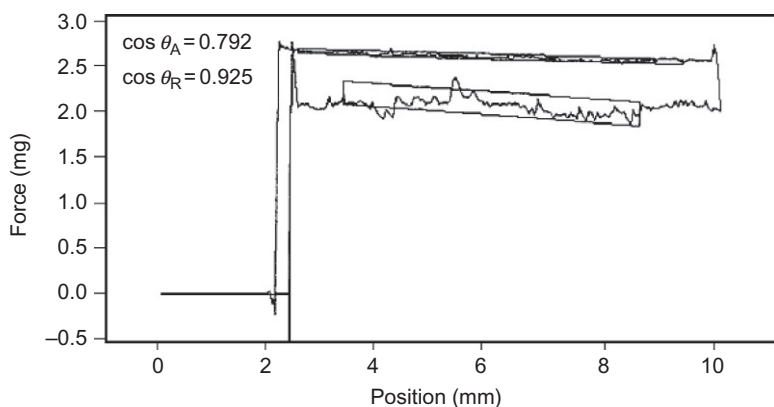


FIGURE 10.21 Wetting and dewetting cycle of water on an optical fiber of 125 μm diameter after mechanical stripping, ethanol washing, and HF etching ($\theta_A = 37.6^\circ$, $\theta_R = 22.3^\circ$).¹

10.7 FUTURE DIRECTIONS

The optimization and validation of cleaning procedures to remove contamination from surfaces remain difficult. The required level of cleanliness depends on the application for which the product is manufactured. This level will determine which analytical method is suitable to monitor surface cleanliness. Analytical techniques such as X-ray photoelectron spectroscopy (XPS) and time-of-flight secondary ion mass spectrometry (TOF-SIMS) are extremely powerful. However, they are expensive techniques that operate under vacuum and require a high level of skill in their operation. They are poorly suited to routine quality control procedures. While these techniques may determine surface cleanliness, the ability to measure the kinetics of cleaning or contamination presents an even greater challenge. Few studies have been published in the literature.^{57–60} TOF-SIMS is a relatively new technique for identifying organic contaminants on solid surfaces in a “routine” contamination laboratory environment. However, it is a versatile method that offers a wide field of applications, especially in the semiconductor industry.⁶⁰

Contact angle measurements provide a powerful analysis tool. They are able to monitor small levels of contaminants on solid surfaces in a rapid and non-destructive manner. They allow the removal rate of organic contaminants and recontamination of the surface to be determined with ease. They provide an easy tool for the routine control of surface contamination on metals, glasses, ceramics, and other inorganic oxide surfaces. Although they give no direct information on the chemical composition of the contaminant, however, correlating these measurements with TOF-SIMS data can provide an accurate characterization of the contaminants.

Using DCA measurements, probed by a liquid advancing or receding at a finite speed across a solid substrate, Davies et al.⁶¹ examined the removal

kinetics of proteins immobilized on a range of solid surfaces, including glass, stainless steel, and polymeric substrates. The substrates coated with proteins were immersed in a cleaning solution of sodium hydroxide and Triton X-100 detergent. Protein removal kinetics was deduced from analysis of the DCA adsorption profile. The results were used to determine optimum soak times in the cleaning solution. Glass and silicon substrates were cleaned in a few minutes, while stainless steel required cleaning times of the order of 1 h.

Testing surface cleanliness by measuring wettability toward water may be misleading if the surface is contaminated with hydrophilic impurities. For example, if the surface is covered with a layer of surfactant or an inorganic salt, water will spread on the contaminants and this may be incorrectly interpreted as a clean surface. For testing the cleanliness of low-energy surfaces, such as polymers, wettability measurements can be used but in this case one needs a table of values of contact angles on clean polymers (as a reference), which is usually available in the literature (e.g., see Ref. 18).

Contact angle measurements offer a quick and simple test for hydrophobic organic contaminants. The ASTM standard recommends this method to detect hydrophobic organic contamination on glass.⁶² This technique can be complemented with highly sensitive surface analytical techniques such as TOF-SIMS to identify the chemical nature of contaminants. It is also anticipated that 3D standard reference objects will become available as a validation tool and a training aid.³⁵

10.8 SUMMARY

Surfaces, especially metal oxide surfaces, are naturally high energy and easy to coat or bond as long as they are free from low-energy hydrocarbon contamination. Detecting contamination is a critical process step in many industries.

Water is a unique probe because of its large surface tension, which comes from its large capacity to hydrogen bond. Oxide surfaces are a perfect complement to water, and thus water is an extremely sensitive probe liquid. Submonolayer hydrophobic contamination can easily be detected by measuring the contact angle of water on an oxide surface. This is extremely important if the surface is to be coated or bonded in a subsequent process step. A twist on a familiar phrase makes the point: “Cleanliness may be next to godliness, but it sure precedes adhesion.”

Contact angle is a very beneficial technique and is available to any shop or laboratory. It does not require much outlay of capital and is affectionately called a “poor man’s” technique. It does not require a Ph.D. to decipher the results, making it essentially “idiot proof.” It does not make sense to spend orders of magnitude more money on vacuum techniques that do not yield the relevant data that come from wettability measurements. Many of the vacuum techniques probe below the surface, whereas wettability does not. This makes contact angle the most surface-sensitive technique available.

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